

**LIGHTING SCIENCE:
MEASUREMENT OF LIGHT –
THE SCIENCE OF PHOTOMETRY**
AN AMERICAN NATIONAL STANDARD

ANSI/IES LS-4-20(R23)
+ERRATA 1

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THE SCIENCE OF PHOTOMETRY**
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Publication of this Committee report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by the
IES Testing Procedures Committee**



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ANSI/IES LS-4-20(R23) Errata

If you, as a user of ANSI/IES LS-4-20(R23), believe you have located an error not covered by the following revisions, you should e-mail your information to Pat McGillicuddy, pmcgillicuddy@ies.org, or send a letter to Pat McGillicuddy, Senior Manager of Technical Content, IES, 85 Broad Street, 17th Floor, New York, NY 10004.

Please confine your comments to specific typographical errors or misstatements of fact in the document's text and/or graphics. Do not attempt revisions of ANSI/IES LS-4-20.

Additions will be posted to this list online as they become available. This errata list was last updated
9/5/25

Additions are shown in **green highlight**. Deletions are shown with **yellow highlight**.

In the Section 15.1 Interior Measurements, Equation 15-4 has errors:

$$\overline{E} = \frac{R(N-1)(M-1)+QN+T(M-1)+P}{M(N-1)} \quad (15-4)$$

The correct equation should be:

$$\overline{E} = \frac{RN(M-1)+QN+T(M-1)+P}{M(N+1)} \quad (15-4)$$

In Section 15.1 Interior Measurements, Equation 15-6 and its definitions have errors:

$$\overline{E} = \frac{R(L-8)(W-8)+8Q(L-8)+8T(W-8)+64P}{WL} \quad (15-6)$$

where:

W = Number of luminaires per row

L = Number of rows
 R = Average of measurements at stations r-1 through r-4, located at random in the central portion of the area

Q = Average of measurements at stations q-1 and q-2, located 0.6 m (2 ft) from the long walls and at random lengthwise in the room

T = Average of measurements at stations t-1 and t-2, located 0.6 m (2 ft) from the short walls and at random crosswise in the room

P = Average of measurements at stations p-1 and p-2, located at diagonally opposite corners and 0.6 m (2 ft) from each wall

The correct equation should be:

$$\overline{E} = \frac{R(L-2X)(W-2X)+2QX(L-2X)+2TX(W-2X)+4X^2P}{WL} \quad (15-6)$$

where:

L = Length of room

W = Width of room

X = A small distance, such as 1.2 m (4 ft), consistent around the entire perimeter

R = Average of measurements at stations r-1 through r-4, located at random in the central portion of the area

Q = Average of measurements at stations q-1 and q-2, located 0.6 m (2 ft) from the long walls and at random lengthwise in the room

T = Average of measurements at stations t-1 and t-2, located 0.6 m (2 ft) from the short walls and at random crosswise in the room

P = Average of measurements at stations p-1 and p-2, located at diagonally opposite corners and 0.6 m (2 ft) from each wall

Note: If illuminance is measured in lux (footcandles), distances are in meters (feet).

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***When you can measure what you are
speaking about, and express it in numbers,
you know something about it.***

– Lord Kelvin 1883, engineer and mathematical physicist

1.0 Introduction and Scope

1.1 Introduction

Lighting is anchored to meaningful visual phenomena by way of the definition of *light* it adopts: the joining of radiant power, a physical quantity, with visual response, a psychophysical quantity. The utility of the definition allows light to be measured and calculated; that is, light has the aspects of *quantity* that permit the engineering activities of measurement and prediction by calculation. This analytic aspect of light and lighting allows successful experience to be recorded and become quantity recommendations for other lighting projects, allows lighting equipment to be characterized in ways useful to designers, and allows predictions of likely outcomes of proposed lighting designs.

Optical radiation generally refers to all radiation that can be measured using certain techniques and equipment (mirrors, lenses, filters, diffraction gratings, prisms). Thus, visible, ultraviolet (UV), and infrared (IR) radiation are collectively considered as optical radiation.

The measurement of optical radiation, called *radiometry*, is the science of measuring radiant quantities and is part of the general science of measurement, metrology.

Photometry, a special branch of radiometry, is the measurement of radiation accounting for human visual response. The International Commission on Illumination (CIE) defines a “standard observer” in part by the photopic luminous efficiency function of wavelength, $V(\lambda)$. The definition of this standard observer quantifies this visual response and defines the spectral response that photometric measurement equipment needs to exhibit.¹

This standard observer response curve is used as a weighting function applied to the spectral power distribution (SPD) of the optical radiation being

Sidebar: A Historical Note

Photometry is a word first used by Johann Heinrich Lambert as the title to his 1760 Latin treatise on the measurement of light. He coined it by combining the Greek words for *light* ($\phi\omega\varsigma$) and *measure* ($\mu\epsilon\tau\omicron\nu$). Lambert’s word soon found its way into European languages.

measured. The summation across all wavelengths of the weighted SPD defines luminous flux.¹ The weighting and summation are the very core of photometry. Though it is globally accepted and used, $V(\lambda)$ is a compromise that always assumes the same predictable correlation of physical measurements with visual response. However, there are circumstances where photometric quantities are poor predictors of visual response. (Refer to ANSI/IES LS-8-20, *Lighting Science: Vision – Perceptions and Performance*.²) Thus, a basic understanding of photometry is essential to the balance that needs to be struck by a lighting designer between measurement on one hand and visual experience on the other.

Photometry and radiometry are used to determine properties of lighting equipment and materials, as well as aspects of the performance of lighting systems. Some of these measurements require photometric standards (for either sources or detectors) and are usually performed in a photometric laboratory; some are accomplished with equipment designed for field use. **Table 1-1** shows the most common types of photometric and radiometric measurement, along with the equipment and usual place of measurement.

1.2 Scope

This Lighting Science (LS) document describes the various types of photometry and photometric instrumentation, including laboratory and field equipment and measurement types, and instructions for some types of field measurements. However, it does not provide instructions or methodology for performing laboratory tests. For that kind of information, the reader is referred to the IES Lighting Measurement (LM) series of documents.

Table 1-1. Radiometric and Photometric Measurements

	Characteristic	Dimensional Unit	Equipment	Technique
Light	Wavelength	nanometer	Spectrometer	Laboratory
	Color	none	Spectrophotometer and colorimeter	Laboratory
Light Sources	Energy radiated	joules per square meter	Calibrated radiometer	Laboratory or field
	Color temperature	kelvin	Colorimeter or filtered radiometer	Laboratory or field
	Luminous intensity distribution	candela	Photometer and goniometer	Laboratory
	Luminance	candelas per unit area	Photometer or luminance meter	Laboratory
	Spectral power distribution	watts per nanometer	Spectroradiometer	Laboratory or field
	Power consumption	watt	Watt meter or voltmeter and ammeter	Laboratory or field
	Total lumen output	lumen	Integrating sphere or photometer and goniometer	Laboratory
	Lumen distribution	lumen	photometer and goniometer	Laboratory
Lighting Condition	Illuminance	lumens per unit area	Illuminance meter	Laboratory or field
	Luminance	candelas per unit area	Luminance meter	Laboratory or field
Materials	Reflectance	percent	Reflectometer	Laboratory or field
	Transmittance	percent	Transmitometer	Laboratory or field
	Spectral reflectance	percent	Spectrophotometer	Laboratory
	Spectral transmittance	percent	Spectrophotometer	Laboratory
	Bidirectional reflectance	inverse steradian	Luminance meter and goniometer	Laboratory
	Bidirectional transmittance	inverse steradian	Luminance meter and goniometer	Laboratory

2.0 Traceability, Accreditation, and Photometric Standards

2.1 Traceability and Accreditation

Contractual agreements often require that measurements made by laboratories be traceable to national and international measurement standards and that the laboratories be able to support claims of traceability with appropriate documentation and records of equipment used in the measurement process. Such agreements are becoming more common with increasing awareness and adoption of documented

quality management systems based on the ISO 9000 series of standards, which states:

4.11.2b: [The supplier shall:] identify all inspection, measuring, and test equipment that can affect product quality, and calibrate and adjust them at prescribed intervals, or prior to use, against certified equipment having a known valid relationship to internationally or nationally recognized standards. Where no such standards exist, the basis for calibration shall be documented.

The requirement for traceability³ involves the ability to relate individual measurement results, with a stated uncertainty, through an unbroken chain of comparisons to a stated reference source. In this way, the reference source is transferred from the national standards through to the end user of the national measurement system.

To assure the end user of traceability and competence throughout the chain, laboratories may seek recognition through assessment and accreditation by a recognized accreditation body. The assessments and general requirements for calibration or testing laboratories are established by the International Organization for Standardization and the International Electrotechnical Commission.⁴ Laboratories that comply with this guide also comply, by definition, with the requirements of ISO 9002 for the scope of the testing and calibration services covered by their quality management system. The reverse is not true since ISO 9002 registration gives no recognition of the quality of the laboratory's specific measurements. In the United States, a number of federal and private agencies offer calibration laboratory assessment and accreditation services. (In the U.S., contact the National Institute of Standards and Technology [NIST] for additional information. In Canada, contact the National Research Council.)

2.2 Photometric Standards

Photometric standards are objects or detectors designed to provide a uniform basis for all photometric measurement, and are important for several practical reasons:

- Standards permit fair and competitive comparisons between lighting equipment performance, based on photometric measurements, regardless of the place of manufacture or final use.
- Standards permit the expectation of a reasonable correlation between predicted performance of lighting equipment (as determined in the laboratory) and that performance observed in application (as measured with field measurement equipment).
- Standards permit private and government laboratories to calibrate photometric measurement equipment, to evaluate lighting products, and to

guide the development and application of new source and material technologies in lighting.

2.2.1 Types of Standards. International metrology vocabulary distinguishes between types of standards based on quality, importance, and intended use.⁵ The standards described in this section are based on that vocabulary.

Primary Standards:

A primary standard is a standard that is designated or widely acknowledged as having the highest metrological quantities and whose value is accepted without reference to other standards of the same quantity. The candela, maintained by the International Bureau of Weights and Measures (BIPM), is a primary standard.

National (Measurement) Standards:

National standards that define radiometric and photometric quantities are maintained by national standard laboratories.⁶ These standards typically are developed from international standards through a specified, usually complex, experimental procedure. In North America, measurement standards for lighting, including the candela, are maintained by the National Institute of Standards and Technology (NIST) in the United States, the National Research Council (NRC) in Canada, and the Centro Nacional de Metrologia in Mexico. National measurement standards are not directly accessible by other laboratories.

Transfer Standards:

Transfer standards are necessary to link the measurement systems of one laboratory to another (for example, a national measurement laboratory and an industrial laboratory).

They are defined simply as intermediaries used to compare standards. They can be called *traveling standards* when intended for transport between different locations.

Reference Standards:

Reference standards are standards having the highest metrological quality available at a given location or in a given organization, from which the measurements made

there are derived. Reference standards can be derived directly from a national measurement standard or from the reference standards of other laboratories in the calibration chain. They usually are prepared with precise electrical and radiometric measurement equipment.

Working Standards:

Working standards are used for routine measurements in a laboratory and are usually prepared and calibrated by that same laboratory from its own reference standard.

Other Terms:

Other nomenclatures have evolved from historical usage but do not represent the internationally accepted definitions, and they are not all consistent. For example, the term *primary standard* is often used to designate a standard source that was obtained from a national standards laboratory and that is used only to make other working standards for everyday use in that laboratory. Sometimes, a primary standard is called a *master standard*. The term *secondary standard* is also commonly used in private laboratories to distinguish a standard from the one called primary, and sometimes the terms *secondary standard* and *working standard* are used interchangeably. The term *tertiary standard* is used if there are three levels of standards deployed.

3.0 Visual Photometry

The earliest instruments for measuring luminous quantities depended on visual appraisal.^{7,8} Such methods lacked both precision and accuracy, largely because the results were dependent on the individual observers making the measurement. Even for a particular observer, measurement reproducibility was limited because a number of variables influencing the measurements could not be controlled or explained. These visual methods are now rarely used, having been replaced with photometric measurements made using calibrated physical instruments that respond to optical radiation. However, visual assessment is still a fundamental part of the psychophysical investigation of visual perception, and forms the foundation of the process that leads eventually to quantification of perceptual effects.

4.0 Physical Photometry

The development and standardization of the $V(\lambda)$ function has allowed visual assessment to be replaced with a physical one: radiometric detection, spectrally modified to mimic the $V(\lambda)$ function. Devices with a spectral response like that defined by $V(\lambda)$ provide the basis for physical photometry.

4.1 Types of Detectors

There is a broad range of detectors available, and the best detector for an application depends on the requirements of spectral response, geometry, and quality. The characteristics of the signal, such as signal-to-noise ratio, amplitude, time response, and frequency bandwidth, all influence the suitability of a detector. The detector system's linearity range, field of view, noise equivalent power, and window transmission, as well as other factors, affect the measurements it can reliably make.

4.1.1 Phototubes. A phototube is a vacuum- or gas-filled glass tube containing a photo-emissive surface as the source of electrical current. Photons striking the photo-emissive surface release electrons by the photoelectric effect, and those electrons are collected by an anode having a higher voltage. The most useful form of phototube for photometry is the photomultiplier tube (PMT).

PMTs employ a photocathode, which emits electrons when irradiated. The spectral sensitivity of a photomultiplier tube depends on the entrance window and photocathode material, for which many choices are available. When photons strike the photocathode, electrons are emitted and then accelerated through a series of electron multipliers (dynodes), where the signal is greatly multiplied. The electrons are collected by an anode, where the output current is measured. A voltage divider chain connects the elements in the PMT in such a way that electrons are accelerated from one stage to the next. Typical PMT designs employ up to 15 stages of dynodes and produce signal gains from several thousand to hundreds of millions. The voltage required to operate the PMT can vary from 500 to 2,000 V, depending on the tube construction and number of dynodes. The overall gain of the PMT is controlled by the voltage applied between elements. A high degree

of voltage regulation is required for accurate operation. PMTs offer the highest sensitivity and are used when extremely low amounts of light are measured.

PMT detectors produce an output signal in the absence of light (a “dark current”), due to thermionic emission. The dark current can be reduced by lowering the temperature of the PMT. Most PMTs exhibit gain differences when exposed to magnetic fields or when their orientation in Earth’s magnetic field is changed. Magnetic shielding is required in most applications. Most PMTs are shock sensitive; rough handling can cause failure or loss of previous calibration. All phototubes have highly selective spectral response characteristics. Depending on the photo-emissive cathode material used, a phototube can be used for ultraviolet (UV), visible, or near-infrared (near-IR) measurement; however, a single phototube cannot cover this entire spectral range.

4.1.2 Solid-State Detectors. Solid-state detectors comprise a very large category of detectors incorporating semiconducting materials. All exhibit similar spectral response characteristics; their sensitivity to longer wavelengths increases up to a photon energy limit, where the detector response drops to zero. The useful spectral ranges of solid-state detectors extend from the UV to the far-IR region. Photodetectors may be used in the photovoltaic mode, where the short-circuit current is measured, or in the photoconductive mode, where a reverse-bias voltage is applied and the device is treated as a radiation-sensitive variable resistor.

Silicon photodiodes are commonly used in laboratory and commercial photometers. They offer a broad spectral range and the ability to measure low levels of radiant power. A silicon photodiode is combined with a glass filter to match its spectral response to the $V(\lambda)$ function. Silicon detectors are also used in self-scanning linear arrays, facsimile (fax) machines, spectral measuring instruments, and two-dimensional charge-coupled devices (CCDs).

Photodiodes perform best when operated as current sources into zero-impedance amplifier circuitry. The linearity of silicon photodiodes has been shown to extend over 10 decades with appropriate amplification.

Because very small currents are involved (typically 10^{-13} to 10^{-3} A), proper amplifier design is essential for the performance of these photometric instruments. Test methods, classes, and performance characteristics have been standardized.⁹

A charge-coupled device (CCD) is a linear multichannel array light detector based on complementary metal-oxide semiconductor (CMOS) technology that is capable of capturing a spectral power distribution over a wide range of wavelengths simultaneously. Typical CCD arrays used in spectroscopy may have from 1,024 to 4,096 detectors, which allows measurements to be made at sub-nanometer intervals.

Inside a CCD, incoming light is converted into an electrical charge, which is accumulated over a given integration interval. The accumulated electrical charge is proportional to the intensity of light at a given wavelength.

CCD elements are “leaky”; some charge is accumulated even in the absence of any light, resulting in a dark signal that will need to be subtracted from the data in order for accurate measurements to be made. Since the magnitude of the dark signal increases with temperature, CCD devices often incorporate Peltier coolers or other devices to maintain a stable operating temperature that is below the ambient temperature.

4.2 Detector Spectral Response

The detector is the primary component affecting the spectral response of a radiant-power-measuring instrument. Photomultiplier tubes (PMTs) and silicon photodiodes are the most commonly used detectors in radiometers and photometers. These detectors respond differently to different regions of the spectrum. The spectral range of the detector is matched to the spectral region to be measured. This significantly improves sensitivity and relieves the burden of filtering. Photometers require suppression of UV and IR.

The native relative spectral response of detectors does not match the $V(\lambda)$ function, and so they cannot directly determine photometric quantities. Spectral filtering is used to produce a combined detector-filter response that closely matches the $V(\lambda)$ function.

A measure of the closeness of this match can be calculated using the CIE parameter f_1' . (See **Section 6.1.1 Spectral Correction Error, f_1'** .)

Some instruments are designed to measure CIE tristimulus values and calculate chromaticities. They use detectors that have to be filtered to produce combined detector-filter responses that match the $X(\lambda)$, $Y(\lambda)$, and $Z(\lambda)$ color matching functions.^{10,11}

Spectral response is particularly important when relatively narrow-wavelength band sources are involved, such as the LEDs that radiate saturated colored light.¹²⁻¹⁴

4.3 Environmental Factors

The environment and conditions of use affect detector performance. Temperature, magnetic and electric fields, and pulse or transient effects can change detector sensitivity, noise, and dark current, and can cause drift.

4.3.1 Temperature Effects. Temperature variations affect the performance of all photodetectors. Silicon photodiodes are affected only slightly by temperature; however, problems can arise from the effects of temperature on detector response. The transmission of the spectral correction filters can also be affected by temperature. Hermetically sealed detectors provide protection against the effects of humidity and some insulation against temperature cycling. Care should be taken that the effects of high temperature or temperature cycling do not damage cemented layers of the detector filter.

PMTs are quite temperature sensitive. Both dark current and noise increase at higher temperatures. In addition, spectral response can vary significantly with temperature changes. Thermoelectric temperature control is frequently used to control the dark current, noise, and spectral characteristics of PMTs.

4.3.2 Transient Effects. Silicon photodiodes typically exhibit microsecond rise times and no fatigue. The rise and fall times for most photometers employing silicon photodiodes are usually limited by the amplification circuitry. PMTs have nanosecond rise times but exhibit hysteresis, that is output overshoot or undershoot, requiring from seconds to minutes to adapt to large light-

level changes. Precision radiometers and photometers usually employ PMTs with minimum hysteresis.

4.3.3 Effect of Pulsed or Cyclical Variation of Light.

Electric discharge sources flicker when operated on alternating current (AC) power supplies. Precautions should be taken with regard to the effects of frequency, pulse rate, and pulse width when measuring the luminous properties of lamps.¹⁵ It cannot be assumed that an instrument will treat modulation of a light source in the same way as the human eye would.

It is important that the internal capacitance of the detector and the response time of the amplifier to pulsating signals be considered. Special metering circuitry for the integration of pulsed light is available for the measurement of flashing sources.¹⁶

4.3.4 Magnetic Fields. As previously noted, radiometers and photometers containing PMTs can be affected by strong magnetic fields. Commercial instruments containing PMTs use magnetic shielding adequate to protect them from most ambient magnetic fields; however, it is advisable to keep them away from heavy-duty electrical machinery.

4.3.5 Electrical Interference. With electronic instrumentation, electrical interference can be induced in the leads between the detector and the instrumentation. This effect can be minimized by using filter networks, shielding, grounding, or combinations of the above.

5.0 Absolute, Relative, and Substitution Photometry

The photometric properties of equipment and materials can be absolute or relative, and may be determined directly or by the method of substitution.

5.1 Absolute Photometry

Absolute photometry measures and reports quantities as they are actually produced by the equipment being measured, in whatever state that equipment

might be, or whatever the operating or measurement conditions. No corrections, other than instrument calibration, are used. Measurements are made with instruments calibrated from standards to report absolute photometric units. It is recommended that all LED luminaires be photometered using absolute photometry.¹⁷

5.2 Relative Photometry

Relative photometry scales measurements to some presumed level of lamp output or other per-unit basis. In this case, instruments or detectors need not be calibrated absolutely. Rather, relative measurements of the equipment under test are made. Another (often just one) separate measurement is then made with the same instrument using a source with an assumed output. The relative measurements are then scaled based on this second measurement.

5.3 Substitution Photometry

Substitution photometry is the sequential measurement of a photometric property of a standard and then of the same property of the object being tested, using the same (to the extent possible) measurement instrument and geometry. The instrument does not have to be calibrated in absolute units. Knowing the photometric property of the standard and the ratio of the two measurements enables determination the photometric property of the object being tested. Luminous flux, intensity, reflectance, and transmittance are often measured by substitution photometry.

6.0 Instrument Accuracy

Instruments for photometric and radiometric measurement are defined by their application. An instrument can be used as a standalone system, such as an illuminance or luminance meter, or combined with auxiliary equipment such as an integrating sphere to form a lamp-flux measurement photometer. Instruments exhibit a considerable range in precision and accuracy, from custom-built equipment in a national metrology laboratory to commercially available, inexpensive, portable meters used for field measurements. The most common types of instruments are:

- Spectroradiometers
- Illuminance meters
- Spot and image luminance meters
- Integrating spheres
- Distribution goniophotometers
- Reflectometers

Of these instruments, spectroradiometers, integrating spheres, and distribution goniophotometers are specialized instruments usually used in a laboratory. The others are much more common and are often used by lighting professionals in various aspects of their work. Accuracy assessments have been developed for these instruments, and occasionally a comparison survey is conducted and reported.^{5,8,9} The accuracy measures important for these instruments are described in **Section 6.1.1** through **Section 6.14**, and are designated by the CIE as f_1 through f_5 . Other factors have been defined but are less common.

6.1 Factors for All Instruments

Some factors affecting accuracy are common to all photometric instruments. Those that have been standardized are spectral correction, linearity, display error, and fatigue. These are designated f_1' , f_3 , f_4 , and f_5 , respectively.

6.1.1 Spectral Correction Error, f_1' . Spectral correction error, f_1' , is an error determined with respect to CIE Standard Illuminant A (a blackbody radiator at 2856 K). The f_1' is evaluated by adding the absolute values of the deviation of the detector's relative spectral responsivity from the $V(\lambda)$ function. That is, if a detector is more sensitive in the blue and less sensitive in the red than the $V(\lambda)$ curve, the respective positive and negative errors do not cancel out when summed. CIE S 023/E2013 characterizes f_1' as "the degree to which the relative spectral responsivity curve $s(\lambda)_{\text{rel}}$ [of the detector] matches the spectral luminous efficacy curve $V(\lambda)$ of the human eye for photopic vision."¹⁸ The spectral correction error f_1' is defined as:

$$f_1' = \frac{\int_{\lambda} |s^*(\lambda)_{\text{rel}} - V(\lambda)| d\lambda}{\int_{\lambda} V(\lambda) d\lambda} \times 100, \quad (6-1)$$

where:

$s^*(\lambda)_{\text{rel}}$ = normalized relative spectral responsivity of the detector

The normalized relative spectral responsivity of the detector is determined by an assessment that compares it to CIE Illuminant A:

$$s^*(\lambda)_{\text{rel}} = \frac{\int_{\lambda} S(\lambda)_A V(\lambda) d\lambda}{\int_{\lambda} S(\lambda)_A s(\lambda)_{\text{rel}} d\lambda} \times s(\lambda)_{\text{rel}}, \quad (6-2)$$

where:

$S(\lambda)_A$ = spectral distribution of illuminant A

$s(\lambda)_{\text{rel}}$ = relative spectral responsivity of the detector

$V(\lambda)$ = photopic luminous efficiency function of wavelength

6.1.2 Linearity Error, f_3 . The linearity of a detector is a property describing how constant the ratio of light input to detector output is over a measuring range of the photometer. An illuminance meter that measures from 0.1 to 100 lx could have three such measuring ranges: 0.1 to 1 lx, 1 to 10 lx, and 10 to 100 lx. For each range of measurement, the error term $f_3(Y)$ is calculated; the largest of the three terms is then given as the nonlinearity error, f_3 , for the photometer.

Generally, most detectors are linear over a specific range and become nonlinear outside set limits. That range should be stated. In addition, the linearity of a detector may be affected by the electronic circuitry to which the output is being fed.

For each range of the photometer, the nonlinearity is characterized by linearity error f_3 , defined as:

$$f_3(Y) = \left(\frac{Y_{\text{output}} X_{\text{limit}}}{Y_{\text{limit}} X} - 1 \right) \times 100, \quad (6-3)$$

where:

$f_3(Y)$ = nonlinearity of a specific range

X_{limit} = maximum illuminance level of the range

X = (typically) 1/10 maximum illuminance level of the range

Y_{output} = photometer reading for input X

Y_{limit} = photometer reading for input X_{limit}

Then,

$$f_3 = f_{3Y_{\text{max}}}$$

6.1.3 Display Error, f_4 . For digital meters, display uncertainty (usually ± 1 digit), the maximum value of the

display (1,999 for a 3½-digit display), and the analog-to-digital (A/D) converter error are considered. Display error f_4 is defined as:

$$f_4 = \left(|f_{\text{display}}| + \frac{|kd|}{|P_{\text{max}}|} \right) \times 100, \quad (6-4)$$

where:

f_{display} = A/D readout display error (from manufacturer)

k = range change factor (that is, 10 for one decade)

d = display uncertainty

P_{max} = maximum value of the display

6.1.4 Fatigue Error, f_5 . Fatigue is a change in the detector's responsivity, usually decreasing with higher levels of incident light but recovering at lower levels. Fatigue is prominent in selenium detectors. Silicon photodiodes normally do not have fatigue other than by temperature effects. The changes are temporal and reversible. Other factors are at work when considering fatigue:

- Spectral responsivity may change.
- Detector heads suffer from temperature effects as well as fatigue when irradiated at high levels.
- Thermostatic control of the detector head does not necessarily eliminate fatigue or temperature effects.

To calculate f_5 , the detector first should be kept in the dark for 24 hours before the test. The photometer is then set at a distance from a source (stabilized illuminant A) such that the maximum allowable level of illuminance is falling on the sensor. Detector output measurements are then made after 10 seconds and 30 minutes. Fatigue error f_5 is defined as:

$$f_5 = \left(\frac{Y_{30 \text{ min}}}{Y_{10 \text{ s}}} - 1 \right) \times 100 \quad (6-5)$$

where:

$Y_{30 \text{ min}}$ = detector output after 30 minutes

$Y_{10 \text{ s}}$ = detector output after 10 seconds

6.2 Illuminance Meter Cosine Response Error, f_2 .

Cosine response means that the detector's output is in direct proportion to the cosine of the angle at which the optical radiation is incident on the photometer head. Standard illuminance measurements at a plane

are typically made using a detector head having cosine response. With the detector placed in front of a stable point source, the procedure for calculating f_2 is a matter of rotating the detector from 0 to 85 degrees, measured with respect to the normal to the face of the detector at 0 degrees, and recording data at, say, 5-degree intervals. The cosine-correction error f_2 is defined as:

$$f_2 = \sum_{\theta=0^\circ}^{85^\circ} \left| \frac{Y(\theta)}{Y(0^\circ)\cos\theta} - 1 \right| \times 100, \quad (6-6)$$

where:

$Y(\theta)$ = signal output as a function of angle of incidence

$Y(0^\circ)$ = signal output at normal incidence

θ = angle of incidence measured from the perpendicular to the detector plane

Note: Angular increments depend on the precision of determination of f_2 .

6.3 Luminance Meter Surround Field Error, $f_2(u)$

Luminance meters are designed to measure light within a specified acceptance angle. However, no optical system can eliminate all stray light, or “flare,” outside that acceptance angle. The stray light splashing into the acceptance angle represents another source of error. The metric $f_2(u)$ is a measure of how well the luminance meter baffles light from outside the acceptance angle. The method for determining this error involves a measurement with and without a gloss trap (opaque, diffuse, black material) in front of a uniform luminance source that is at least 10 times as large as the acceptance area. The gloss trap is 10 percent larger than the acceptance angle. The surround field error $f_2(u)$ is defined as:

$$f_2(u) = \frac{Y_{(\text{surround})}}{Y_{(\text{total})} - Y_{(\text{surround})}} \times 100, \quad (6-7)$$

where:

$Y_{(\text{surround})}$ = detector output for measurement with gloss trap in place

$Y_{(\text{total})}$ = detector output of total field

For luminance meters, detector response is also characterized by $f_2(g)$, which defines the uniformity of response of the detector (i.e., its spatial symmetry). Some areas of the detector surface may be more or less sensitive than others are.

7.0 Measuring Spectra

Spectral measurements are assessments made that account for the wavelength of the optical radiation involved and are among the most fundamental measurements that can be made of the optical radiation from light sources and of the optical properties of materials that interact with optical radiation.

There are different types of spectral measuring systems to suit specific applications, but they all generally incorporate the following elements:

- Collection optics to receive and limit the radiation to be measured
- A monochromator that disperses radiation coming through an entrance slit and selects a narrow range of wavelengths that it sends to a detector
- A detector or detector array
- Electronics to process the detector or array signal
- Some type of display and electronic data output connection

Altogether, such a system is called a *spectroradiometer*.

The monochromator houses a dispersing element, often a diffraction grating, which separates the various wavelengths of the input spectrum. The monochromator has an entrance aperture, usually in the form of a rectangular slit, through which the collected radiation enters; perhaps some optical elements that image the entrance slit onto the dispersing element(s); and an exit slit through which selected wavelengths of the dispersed radiation pass.

In most spectroradiometers, an array of irradiance detectors is positioned at the exit slit to measure the radiant power of the source at wavelengths throughout the optical spectrum.

Detector array spectrometers acquire spectral data simultaneously without mechanical moving parts. In modern instruments, photodiode arrays are used, in which each diode element, or line of diode elements, detects incident radiation in a narrow wavelength band, responding to the irradiance produced by the monochromator in

that band. The wavelength resolution depends on the sensitivity of the detector and the narrowness of the bands.¹⁹ High-dispersion gratings and narrow array elements permit spectral resolutions to $\frac{1}{2}$ nm.

Silicon photodiodes do not exhibit a spectrally flat response, and so array detector spectroradiometers have internal data processing capabilities to account for the varying sensitivity of the array diode elements through the optical spectrum. Other calibration parameters are handled electronically as well. Electronically scanned silicon photodiode arrays provide nearly instantaneous determination of a spectral power distribution.²⁰

Figure 7-1 shows a schematic diagram of a diffraction spectroradiometer and array detector.

In some high precision, high sensitivity instruments, the detector is a photomultiplier tube that is fixed at the exit slit of the monochromator. The spectrally dispersed output of the monochromator is swept across the exit slit by mechanically rotating the dispersion element. Industry guidelines and standards exist for spectroradiometric measurements, covering instrumentation, calibration and measurement procedures, and data reduction.²¹⁻²³

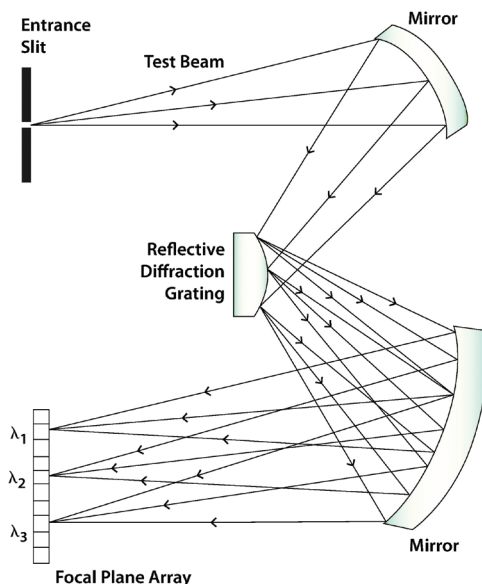


Figure 7-1. Spectroradiometer. Schematic diagram of a spectroradiometer using mirrors and a diffraction grating as a monochromator, and an array of silicon photodiodes as the full-spectrum detector. Some implementations use only one mirror; others use a lens for spreading or focusing. (© Illuminating Engineering Society)

7.1 Using Spectroradiometers

Spectroradiometers are used in several ways:

- Measure relative and absolute spectral power distribution (SPD) of sources, spectral radiance and irradiance, spectral reflectance and transmittance, and spectral scattering.²⁴ In each case, the basic components are those outlined in **Section 7.0**, the difference being how radiation is sampled and what type of spectral comparison standard is required.
- Measure the relative spectral responsivity of detectors. Radiometric detectors should exhibit a spectrally flat response over a stated range of wavelengths. Photometric detectors should exhibit a spectral response defined by the photopic luminous efficiency function. In both cases, measurement equipment can be tested and/or components designed using spectroradiometric measurements.
- Used as spectrophotometers. In this case, the spectral response is not flat but should be the photopic luminous efficiency function. In most modern spectroradiometers, this conversion from radiometric to photometric assessment is performed electronically; no adjustment or filtering is performed on the flux falling on the array detector. Though called a spectrophotometer because of its principal application to measurements in the visible spectrum, this type of instrument is often designed for measuring UV and near-IR radiation. Some spectrophotometers are designed specifically for UV or IR measurements.

The reduction in cost and size of electronic components and the increase in precision and reliability of diffraction gratings have permitted the development of very compact spectroradiometers. Portable spectroradiometers for determining spectral radiance and luminance are now common.

7.1.1 Measurement of Spectral Power Distribution

(SPD). A spectroradiometer is used to measure the SPD of light sources. Two methods are usually employed. In the first, the collection optics of the system directs optical radiation into a small integrating sphere or diffuser plate positioned in front of the entrance slit of the monochromator. This geometry is typically used to

measure the spectral irradiance of a light source. In the second method, the optics of the system directs radiance from a uniform-source integrating sphere, an irradiated highly reflecting diffuse target, or a diffusely emitting lamp into the entrance slit of the monochromator.

By convention, relative SPD graphs are normalized so that their peak power is equal to 100, though in some cases relative SPD graphs are normalized so that the value at 560 nm is equal to 100.²⁵ The wavelength bandwidth depends on the instrument used and the requirements of the data.

7.1.2 Spectral Reflectance and Transmittance.

A spectrophotometer is used to determine spectral reflectance and transmittance properties of materials. In some reflectance instruments, standards are used to permit comparison of optical radiation reflected from a standard to that of a measurement sample through individual wavelength bands.

Spectral transmittance is often done by splitting the beam of narrow-band optical radiation from a monochromator into two and passing one through the sample to be measured. Comparison of the two beams reveals the absorption of the sample in that narrow wavelength band.

Spectral reflectance may vary with incident and exitant directions. Standards have been developed for spectral reflectance and transmittance measurement conditions and procedures,²⁶ specifically when these quantities are used in determining object color and to provide a means of examining the color of a material for analysis, standardization, and specification.^{11,27}

8.0 Measuring Illuminance

Because illuminance recommendations are a common and important part of lighting design, illuminance is the most common type of photometric measurement made. These measurements are also common because illuminance is conceptually simple. Illuminance measurements are used to verify lighting system performance and achievement of design goals.

8.1 Illuminance Meters

Since illuminance measurements are relatively common, commercial illuminance meters are available in a wide range of quality and capability. The simplest illuminance meters consist of a photodiode with a photopic correction filter. The photodiode is connected to an operational amplifier with a display. These can be bench top, rack mountable, or portable. They can be enclosed in one case, or, as is more common with laboratory photometers, the detector and filter can be in one module that is connected by a cable to a console, at a convenient distance, containing the amplifier and display. The electrical scheme can be anything from a simple amplifier with manual controls to a programmed microprocessor with routines for calibration, measurement, and conversion of display units. Some meters include communication ports for remote operation and data manipulation. Examples of commercial instruments are shown in **Figure 8-1**.



Figure 8-1. Examples of commercial handheld illuminance meters: integral detector head (right) and external remote detector (left). (Image © Konica Minolta)

8.2 Angular Response

By definition, illuminance meters should exhibit a response to incident flux that decreases as the cosine of the angle of incidence. Flat detectors and response-correcting filters do not have this type of response, and therefore the detector of an illuminance meter is usually specially configured to come close to the so-called cosine response. Detectors used in most illuminance

photometers now have diffusing covers or some means of correcting the readings to a true cosine response. Solutions to the cosine problem include placing over the detector a flashed opal glass, a diffusing acrylic disk, or an integrating sphere with a knife-edge entrance port. However, with flashed opal glass or a diffusing acrylic disk, at high angles of incidence light will reflect specularly, so that the readings remain too low.

This can be compensated for by allowing light to enter through the edges of the diffuser. The readings at very high angles will then be too high but can be corrected by using a screening ring. The addition of auxiliary optics to improve cosine response can affect the photometric and directional response.⁵

Illuminance meters of low quality can exhibit inaccurate responses at large angles from the normal, with errors as much as 25% below the true illuminance value. This can be important when making illuminance measurements in a space with strong sidelighting from daylight. Measurements of illuminance produced by light from high incidence angles (measured from the normal to the meter) can be in serious error if the illuminance meter has poor spatial response correction at high angles. Similar problems can arise in measurements of roadway and sports field illuminance from distant luminaires.

Even with good spatial response correction, it is critical that the illuminance meter head be level or accurately parallel to the intended measurement plane. This is particularly important during field measurements of lighting systems where light is received from one or a small number of discrete sources, such as in roadway lighting. Instruments are available in which the detector is gimbal-mounted and self-leveling. This removes problems in trying to measure horizontal illuminance on uneven or sloping surfaces.

8.3 Spectral Response

Most illuminance meters alter the spectral response of the detector element with specially constructed spectral filters that when combined with the detector produce a system with a spectral response that approaches the $V(\lambda)$ function. (See **Section 4.2 Detector Spectral Response**.)

9.0 Measuring Luminous Intensity

Luminous intensity (I) is usually determined in an indirect manner: a value of intensity is inferred using the inverse-square cosine law, an illuminance measurement, and the distance at which the illuminance measurement is made. That is, the effective luminous intensity is determined:

$$I = E/D^2, \quad (9-1)$$

where:

E = illuminance measured as produced by the source

D = test distance from the source to the plane of the illuminance measurement

This assumes that the plane of the illuminance measurement is perpendicular to a line from the source to the illuminance detector, and that the source is essentially a luminous point. For large sources this process gives the *equivalent luminous intensity*.

In most cases, far-field luminous intensity is determined, and this requires a test distance significantly larger than the largest luminous dimension of the source. For architectural luminaires, this can involve test distances greater than 8 m (25 ft).

9.1 Optical Bench Photometry

Optical bench photometers are used for the calibration of instruments for illuminance measurement. Optical bench photometers provide a means for mounting sources and detectors in proper alignment and a means for easily determining the relative distances between them. If the source of unknown luminous intensity is distant enough from a calibrated detector so that its radiation can be treated spatially as if it were emanating from a point, the inverse-square law can be used to compute the unknown intensity from the illuminance determined by the calibrated detector. This is a determination of absolute *luminous intensity*.

9.2 Distribution Photometry

A series of luminous intensity measurements around a source characterize its intensity distribution. These measurements are made with a combined photometer

and goniometer, usually referred to as a *goniophotometer*. The source can be a lamp or a luminaire. The intensity is determined at a series of positions around the source at a set of angles spanning directions appropriate for the source, and at sufficiently small intervals to provide a density of information consistent with the intended use of the data and the nature of the source. The coordinate system used for these angles depends on the construction of the goniophotometer and the types of sources it is intended to measure.

There are three common types of angular coordinate systems used in distribution photometry: Type A, Type B, and Type C.²⁸ As a practical matter, most distribution photometers involve the coordinate system of Type C photometry, and if photometric data are required in Type B form, it is provided by interpolating the Type C data. The Type A system is widely used for automotive device photometry.

Distribution photometers for relatively small sources require concomitantly small test distances, and the instruments can be relatively compact and fit into small laboratory spaces. Architectural lighting equipment is often large and requires large test distances. For this reason, the most common type of distribution photometry used for testing architectural lighting equipment is the *moving-mirror goniophotometer*.

In this instrument, the lamp or luminaire has a fixed position and rotates only about a vertical axis. This keeps the lamp or luminaire position with respect to gravity and thermal air motion fixed. An illuminance detector is in another fixed position. A mirror rotates about the lamp or luminaire to one of the required distribution directions, effectively “folding” the optical path from the lamp or luminaire to the fixed detector, which permits the positions of source and detector to remain fixed. This in turn allows large test distances can be accommodated in relatively small spaces. **Figure 9-1** shows a commercial moving-mirror type of goniophotometer. In this photograph, the operator is adjusting a luminaire on the mount, which is lowered for servicing. During measurements, the mount is raised along the vertical track to bring the photometric center of the luminaire in line with the rotation axis of the mirror.

The use of mirrors can be a source of error if the source being measured produces strongly polarized light, since the mirror reflectance can change significantly with the incident direction of strongly polarized light. Back surface mirrors can also introduce refraction and spatial errors that front surface mirrors do not.²⁹

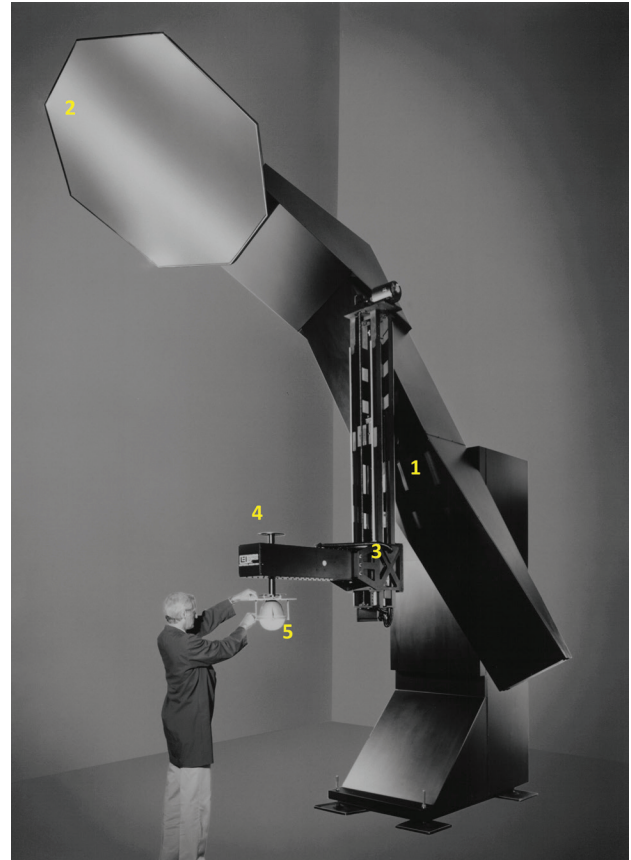


Figure 9-1. An example of a commercial goniophotometer with fixed luminaire position and rotating mirror. (1) Principal mirror rotation axis; (2) mirror; (3) luminaire mount; moves luminaire horizontally and vertically to position luminaire at the intersection of the two centers of rotation of the goniometer, which is the photometric center; (4) azimuthal rotation axis of the luminaire; (5) test luminaire. (Image © Lighting Science)

10.0 Measuring Flux

Luminous flux measurements are used in lamp and luminaire photometry. Lamp lumens, luminaire lumens, and luminaire efficiency are equipment parameters requiring the determination of flux. Flux can be measured directly, as described in this section, or determined

indirectly from a luminous intensity distribution, as described in **Section 14.6.1 Zonal Lumens**.

10.1 Integrating Sphere

The integrating-sphere photometer is used to measure the total luminous flux from a source. The most common type is the Ulbricht sphere.³⁰ The theory of the integrating sphere assumes an empty sphere whose inner surface is perfectly diffusing and of uniform nonselective reflectance. Every point on the inner surface then reflects to every other point, and the illuminance at any point is therefore made up of two components: the flux coming directly from the source and that reflected from other parts of the sphere wall.

With these assumptions, it follows that the illuminance, and hence the luminance, of any part of the wall due to reflected light only is proportional to the total flux from the source, regardless of its distribution. The luminance of a small area of the wall, or the luminance of the outer surface of a uniformly diffuse transmitting window in the wall, when carefully screened from direct light from the source but receiving light from other portions of the sphere, is therefore a relative measurement of the total luminous flux from the source.

Figure 10-1 shows the Ulbricht-type integrating sphere with a high-reflectance, diffuse white interior.



Figure 10-1. Example of a commercial Ulbricht-type integrating sphere with remote source power and measurement display. (Image © Labsphere)

The presence of a source having finite dimensions, its supports and electrical connections, the necessary baffles or shields, auxiliary accessories, and the exit window or ports are all departures from the basic assumptions of the integrating-sphere theory. While durable high-reflectance diffuse material and coatings are now available for sphere interiors, none exhibits the ideal properties of perfect diffusivity and spectral non-selectivity. Despite these limitations, if the reference source and the test source are similar in shape, size, surface reflectance characteristics, and light distribution patterns, the errors introduced by an imperfect integration can be small. For accurate measurements of sources dissimilar from the reference source, corrections need to be applied for self-absorption, spectral mismatch, and spatial non-uniformity, which are inherent with integrating-sphere lamp measurement photometry.^{31,32}

An alternative to the substitution method uses an integrating-sphere with an external source, as shown in **Figure 10-2**.³³ In this geometry, the total luminous flux of a source in the integrating sphere is calibrated against an external reference source calibrated for illuminance, at an aperture outside the integrating sphere. The total luminous flux of the external source can be determined from the known illuminance produced by the external standard and aperture area. Additionally, other detectors can be connected to the exit port of the integrating sphere, creating integrating-sphere spectroradiometer, for example.

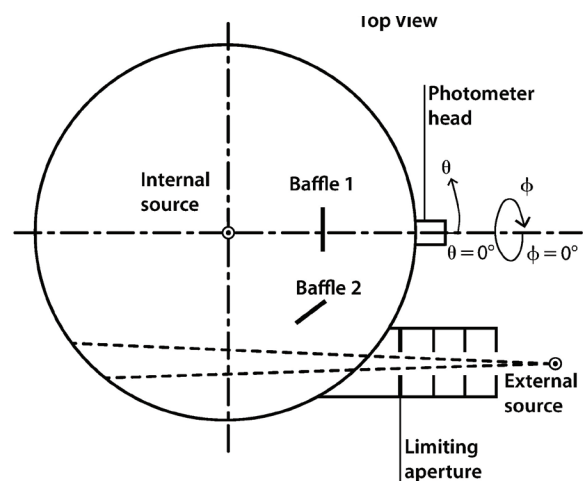


Figure 10-2. Research Standard integrating sphere using an external source of to measure flux from a source inside the sphere. (© Illuminating Engineering Society)

11.0 Measuring Luminance

Luminance meters are essentially illuminance meters with the addition of suitable optics to image an object onto the detector. They operate on the principle embodied in the equation relating illuminance, luminance, and solid angle:

$$E = L \Delta\omega \cos\xi, \quad (11-1)$$

where:

L = object luminance

$\Delta\omega$ = solid angle subtended by the source to the illuminance measurement point

$\cos(\xi)$ = cosine of angle between direction of object solid angle and perpendicular of the illuminance plane

E = illuminance produced by the object

If the plane of the illuminance measurement is perpendicular to the source, **Equation 11-1** can be rearranged as:

$$L = \frac{E}{\Delta\omega} \quad (11-2)$$

That is, luminance can be determined from an illuminance measurement made through a limiting aperture of known solid angle.

In many luminance meters, a means of viewing the object is provided so that the user can see the area that is being measured as well as the surrounding field. Because of the similarity of this optical system to a telescope, these instruments are also called *telephotometers*.

Changing the focal length of the objective lens changes the field of view and the size of the measurement field. Some systems have apertures of various sizes to further define the measured area. Angular measurement fields from seconds of arc to several degrees can be selected.

Typically, modern luminance meters use silicon photodiodes or photomultiplier tubes (PMTs; see **Section 4.1.1**). The amplifier sensitivity may be either manually selected or automatic. Color filters can be incorporated for color measurements, and neutral density filters to extend the dynamic range.

Photodetectors are typically photodiodes for portable and low-sensitivity instruments, and PMTs for high-sensitivity instruments. Most instruments have a sensitivity dynamic range of at least four, and many incorporate attenuation screens or neutral-density filters for additional range. The electrical scheme can be anything from a simple amplifier with manual controls to a programmed microprocessor with routines for calibration, measurement, and conversion of display units. Some meters include communication ports for remote operation and data manipulation.

11.1 Spot Luminance Meters

A common type of luminance meter determines the luminance over a relatively small area, typically subtending 3° or less to the observation point.

11.1.1 Beam Splitter Spot Meter. This type of photometer employs a beam splitter behind the objective lens, which divides the incoming radiation into two paths. Approximately half of the radiation passes through the beam splitter and is focused on an aperture defining the measurement field. The radiation passing through the aperture can be measured with either a PMT or a solid-state detector.

The radiation reflected from the beam splitter is focused on a reticle* having an etched pattern with the same dimensions as the measurement aperture. A viewing system with an eyepiece allows the user to see the field of view and an outline of the area being measured. It is critical that the reticle be carefully aligned with the measuring field. Readings are usually in cd/m^2 or cd/ft^2 . Some instruments may include colorimetric filter options. Field-of-view capabilities may range from 0.25° to 10°, with sensitivity ranging from 10^{-2} to 10^6 cd/m^2 . **Figure 11-1** shows a commercially available beam-splitter spot-luminance meter.

Although good measurements can be made with this type of instrument, it does have some noteworthy disadvantages. Among these are loss of illumination to both the detector and the viewer; introduction of polarization, which affects the measurement

* A *reticle* is a network of fine lines, wires, or the like placed in the focus of the eyepiece of an optical instrument. Source: www.dictionary.com (accessed 2018 Aug 24).

of polarized sources; and the difficulty of changing apertures and reticles for different measurement fields. In general, a low-cost instrument using a beam splitter will provide adequate but not exact location of the actual measured spot.



Figure 11-1. A commercial beam-splitter spot-luminance meter (left) and a commercial aperture mirror spot-luminance meter (right). Both luminance meters require proper focusing to produce accurate measurements.

(Left image © Konica Minolta; right image © Photo Research)

11.1.2 Aperture Mirror Photometers. Most of the problems of the beam-splitter spot meter are addressed by the aperture mirror photometer. There is no beam splitter to introduce polarization error or reduce the brightness at either the measuring aperture or the viewed image. The image formed by the objective lens falls on an angled first surface mirror with a through-hole for the measuring aperture. The viewing optics are focused on the aperture, which appears as a black circle. The field around the measurement aperture is clearly seen in the eyepiece. This arrangement allows apertures to be changed without the need to change precisely aligned reticles as well.

A disadvantage of the aperture mirror photometer is that if a small source is imaged within the measuring aperture, it cannot be seen in the viewing optics. Instruments of this class usually employ high-quality detectors, one or more neutral-density range-multiplying filters, lens options, and some degree of colorimetric capability. They are available with internal microprocessor control and direct reading capability for luminance in several units,

for color chromaticity coordinates, and for correlated color temperature. The full-scale sensitivity for the best laboratory instruments ranges from 10^{-4} to 10^8 cd/m².

Figure 11-1 shows a commercially available aperture mirror spot luminance meter.

11.2 Digital Luminance Meters

Developments in imaging devices have provided a powerful tool for luminance measurements of complete scenes. Cameras equipped with a charge-coupled device (CCD) array are able to capture and digitize electronic images of visual scenes.³⁴⁻³⁶ Providing that the proper controls are applied, the digital image can be used to determine the luminance at every point in the scene, corresponding to the pixels of the camera's CCD array. **Figure 11-2** shows an example of a commercial CCD-based luminance meter.

A complete photometric capture can be carried out and saved. As the information is provided in digital form, complicated functions of luminance images can be analyzed and reported for uniformity, contrast, spatial characteristics, and other photometric values.

Some systems also provide chromaticity values.



Figure 11-2. An example of a commercial imaging luminance meter. (Image © Photo Research)

This form of photometry requires many factors to be controlled in the instrument and software if accurate results are to be obtained. The CCD and optical attachments need to be of high quality. "Field flattening" adjustments are required for all lenses, which spatially distort the image to at least some degree. To measure the complete dynamic range of luminances in most interior scenes, earlier (8-bit) systems required capture of multiple images at different exposure settings. Newer, 16-bit systems are available that cover a dynamic range of over 65,000:1.

Cooling the CCD array may be needed for low luminances to reduce noise and dark current, to increase the detection limit, and to minimize effects of ambient temperature.

Applications for imaging photometers include the energy distribution of lamps (for example, flood lamps and automobile headlights), production line quality control, luminance uniformity of a projected scene, and complicated analyses of scene illumination.

12.0 Measuring Reflectance and Transmittance

Reflectance and transmittance are basic properties of architectural materials and are usually used to express the bulk properties of a material: total flux reflected or transmitted, in ratio to the flux incident.

This section discusses spectrally integrated reflectance and transmittance. (See **Section 7.1.2 Spectral Reflectance and Transmittance** for spectral measurements.)

Generally, reflectance and transmittance are not simply a property of a material. Rather, they also depend on the measurement geometry, i.e., the spatial relationship between the source and the detector. In some cases, bidirectional reflectance or transmittance is measured and specified; in others, spatially integrated values are measured and reported.

12.1 Reflectometers (and Transmittometers)

Reflectometers are reflectance measurement photometers. Reflectance measurements typically fall under three categories: diffuse, specular, and a mix of specular and diffuse reflectances. The design of the reflectometer and the method of measurement depend on the reflectance properties of the sample material and what kind of reflectance one desires to measure.³⁷

The fraction of the incident light reflected can be difficult to determine directly, particularly for diffuse reflection, and so in some cases reflectance is expressed

as a *reflectance factor*: the ratio of the reflectance of a sample to that of a reflectance standard under the same measurement geometry. Three commonly used reflectance standards are a polished front-surface mirror, a polished black glass having a specified index of refraction, and a totally diffuse reflector (for example, barium sulfate[†]).³⁰

In one method commonly used for measuring total reflectance, the sample is illuminated by a narrow cone of light from a given angle, typically 10° or less from the normal to the sample surface, and the reflected light is collected over the entire hemisphere surrounding the sample. Instruments of this type are said to employ a *conical-hemispherical geometry*. **Figure 12-1** shows an example of a commercial reflectometer. The hemispherical flux collection is often accomplished by means of an integrating sphere with a detector, arranged so that the detector does not receive light reflected directly from the sample, but rather views the sphere wall. In this way, the signal is proportional to the total flux reflected from the sample. A similar technique can be used to measure transmittance.



Figure 12-1. An example of a commercial reflectometer. Reflectometers are often designed specifically for surface and finish measurements. This is a solar reflectometer with multiple sources and filters for the measurement of reflectance in the broad spectrum of optical radiation produced by the sun. (Image © Devices and Services)

[†] Barium sulfate, BaSO₄, is a white crystalline solid that is odorless and insoluble in water. It occurs as the mineral barite, which is the main commercial source of barium and materials prepared from it. The white opaque appearance and its high density are exploited in its main applications, which include radiopaque agent, pigment, paper brightener, plastics filler, and in colorimetry as a near-perfect diffuser when measuring light sources. Source: Wikipedia – Barium sulfate (accessed 2018 Aug 24).

The same type of instrument also can be used to measure only that part of the light that is diffusely reflected. One example of a sample that might be measured in this way is one with a very smooth dielectric surface that reflects strongly by means of scattering from pigments or other inclusions beneath the surface. In this case, light specularly reflected from the sample is allowed to escape through a specular subtraction port in the sphere wall, where a light trap can be positioned to absorb the specularly reflected beam.

For measuring color, a *45/0 reflectometer* is often used to evaluate the spectral character of diffusely reflected light. The source and the detector are mounted in a fixed relationship in the same housing. Light is incident on the surface from an angle of 45°, and the detector is positioned above and normal to the sample surface.

Reflectances from plane samples can be measured in many ways. One method employs a reflectometer that compares, with the aid of an auxiliary mirror, the incident flux with the flux after two reflections from the sample. Such a reflectometer is often available as an accessory to commercial spectrophotometers. Another method employs a goniophotometer that allows the user to position the light source and detector at any known angle. In some models, the sample holder can also be repositioned. Applications of the goniophotometer include measurements of gloss, luster, and haze.

Another type of instrument, the *Taylor Baumgartner sphere reflectometer*, measures total reflectance. It consists of an integrating sphere, light source, and a photodiode. The sample is placed at the sample port of the integrating sphere. A collimated beam of light is directed onto the sample from approximately 30° to the normal, and the total reflected light, integrated by the sphere, is measured by the photodiode, mounted in the sphere wall. The collimated light source is then rotated so that the light is incident on the sphere wall, and a second reading is taken. The sample is in place during both measurements, so that the effect on both readings of the small area of the sphere surface it occupies is the same. The ratio of the first reading to the second is the reflectance of the sample for the conditions of the test. Samples of translucent materials should be backed by a light trap.

12.2 Field Measurement of Reflectance

The reflectance of an architectural surface in the field can be determined by the method of substitution and a standard reflectance card. Luminance measurements are made of a spot on, say, a wall. The standard reflectance card is placed over the same spot and its luminance is measured. The wall reflectance at that spot is approximately equal to the reflectance of the standard card times the ratio of the two luminance measurements, as shown in **Equation 12-1**.

$$\rho_{\text{surface}} = \rho_{\text{standard}} \times \frac{L_{\text{surface}}}{L_{\text{standard}}}, \quad (12-1)$$

Though useful, this approximation is subject to large error if the surface being measured is very specular or the illumination at the spot is highly directional.

13.0 Lamp Photometry

Lamp photometry is the measurement of various photometric properties of lamps operating under conditions usually consistent with typical applications. The measured properties can include total emitted lumens, intensity distribution, and spectral power distribution. Since lamps are electrical devices and their photometric properties are sometimes sensitive functions of electrical supply,³⁸ the electrical operating characteristics and conditions of lamps are carefully noted and controlled during testing. Similarly, some lamps have photometric properties that are dependent on their operating temperature, and therefore it is critical that the operating temperature be carefully noted and controlled during testing.³⁹ For these reasons, lamp photometry usually involves a considerable amount of electrical and thermal monitoring, testing, and reporting, in addition to the photometric testing.

Some lamps require auxiliary equipment such as starters, ballasts, or drivers. There is often, though not always, reference versions of this equipment that operate with reference performance and provide reference electrical conditions for the lamp being tested.

13.1 Characterizing Lamps

Lamps, and light sources in general, are characterized by their electrical, radiant, luminous, and life performance properties. (Refer to *ANSI/IES LP-4-20, Lighting Practice: Electric Light Sources – Properties, Selection, and Specification*.⁴⁰) Depending on the lamp type and the intended use of the data, lamp testing can include:

- Electrical properties:
 - Starting, operating, and restrike voltages
 - Electrode voltages
 - Current
 - Power
 - Power factor
- Radiant properties:
 - Total efficiency
 - SPD
 - CRI, CCT, chromaticity coordinates
- Photometric properties:
 - Total flux
 - Efficacy
 - Intensity distribution
 - Beam and field flux
 - Flicker index and other flicker metrics
- Life performance properties:
 - Life
 - Lumen depreciation

13.2 Lamp Testing

Lamp testing is usually intended to describe the photometric, radiometric, electrical, and thermal performance characteristics of a typical individual lamp from a relatively large population of commercially produced lamps of a single type. To the extent that all lamp manufacturing processes, however closely controlled, produce lamps of slightly varying characteristics, lamp testing is performed on a sufficiently large sample of lamps so that the resulting average performance can be reliably used as the rated performance for that lamp type.

Many industry standard lamp types are produced by several different manufacturers, each using different materials, components, processes, procedures, and monitoring and test methods. Thus, lamps of the same type may have systemically different characteristics,

depending on the manufacturer, that are not revealed by nominal lamp data. An example is a systemically different intensity distribution but with nearly the same total lumens.

Radiometric testing of lamps requires equipment beyond the instruments necessary for photometric testing, and though not always the case, making ultraviolet radiation measurements can be hazardous when using sources designed for germicidal applications. Standards for spectroradiometric and ultraviolet radiation testing describe the equipment, safety precautions, and test conditions for these measurements.^{19,40,41}

Most lamp types used in architectural lighting have an appropriate standard governing the equipment, procedures, test conditions, instrumentation, and test report contents to be used or provided in lamp photometric testing. In addition, proper lamp handling and seasoning before testing (except for LEDs) is always part of good practice.⁴² Standards governing all these issues are available for most lamp types. These include:

- Filament⁴³
- Fluorescent⁴⁴
- Compact fluorescent (CFL)⁴⁵
- High intensity discharge (HID)⁴⁶
- Light emitting diode (LED)¹⁷
- Fiber optics⁴⁷
- Low pressure sodium (LPS)

The determination of life performance properties for different lamps has appropriate standards governing equipment, testing conditions, auxiliary equipment, lamp selection and sampling, pre-test procedures, lamp operating cycles, and report content. The types of lamps include:

- Filament⁴⁸
- Fluorescent⁴⁹
- Compact fluorescent (CFL)⁵⁰
- High intensity discharge (HID)⁵¹
- Low pressure sodium (LPS)
- Light emitting diode (LED)⁵²

14.0 Luminaire Photometry

Luminaire photometry is the measurement of the properties of a luminaire, operating under standard test conditions, intended to provide data adequately describing the luminaire's performance and permitting its evaluation as part of the lighting design process. Some data are the direct result of photometric, electric, or thermal measurement; others are determined indirectly, using standard calculation procedures involving the primary data, and provide necessary application information.

14.1 Far-Field and Near-Field Photometry

Virtually all commercial luminaire photometry provides intensity distributions that are far-field. That is, the test distance used is greater than 5 times the largest luminous dimension of the luminaire. Though not usually noted, intensities produced this way are *equivalent intensities*.)

Near-field photometry has several forms, though it is commercially uncommon since its use for lighting calculations requires specialized application software, or special handling when incorporated into standard software.⁵³ Standards exist for near-field photometry of luminaires.⁵⁴

14.2 Absolute Luminaire Photometry

Absolute luminaire photometry involves luminous measurements made with detectors calibrated to provide direct assessment in absolute units. For example, the determination of absolute luminous intensity is made with an illuminance detector calibrated absolutely in lux. Thus, the determination of intensity, I , using the inverted inverse-square law, as in **Equation 9-1**, involves an absolute determination of the test distance, D , in meters and the illuminance, E , produced by the source in lux.

14.3 Relative Luminaire Photometry

Relative luminaire photometry provides an intensity distribution on a per-unit basis.

The basis is an assumed total lumen output of the lamp or lamps usually used in the luminaire. In this case, equivalent luminous intensities are determined

from measurements made with detectors that are not absolutely calibrated. Instead of using **Equation 12-1**, equivalent luminous intensity is determined from:

$$I = \mu k_{\text{scale}} \quad (14-1)$$

In this case, the detector generates an output signal, μ , which is assumed to be proportional to the luminous intensity of the source, which in turn is proportional to the illuminance produced by the source and the square of the distance from the photometric center to the detector. Though not calibrated, it is necessary that the detector exhibit a linear response to the incident flux density. If relative intensities are determined from a series of measurements at a set of angular directions, then in general there will be a different scaling factor, k , for each measurement. If the test distance is constant and other operating factors are unchanged measurement-to-measurement, a single scaling factor can be used for all the data. Then:

$$I(\theta, \psi) = \mu(\theta, \psi) k(\theta, \psi)_{\text{scale}} = \mu(\theta, \psi) k_{\text{scale}}, \quad (14-2)$$

where:

$I(\theta, \psi)$ = equivalent luminous intensity in angular direction (θ, ψ)

$\mu(\theta, \psi)$ = signal from the detector in angular direction (θ, ψ) at a fixed distance k_{scale} for entire set of measurements

After the measurement of a set of relative values, $\mu(\theta, \psi)$, the scale factor, k_{scale} , is determined by isolating the lamps used in the luminaire and assessing their actual (in distinction to their rated) lumen output. The combined effect of photometer calibration and unknown actual lamp lumen output is accounted for in k_{scale} .⁵⁵ Isolating lamps with sensitive thermal properties complicates this process considerably, sometimes making it difficult to determine other derived characteristics from this primary dataset.⁵⁶ For luminaires using LEDs, relative photometry should never be used.¹⁷

14.4 Characterizing Luminaires

Luminaires are characterized by their luminous and application properties. Some luminous properties are measured directly, others derived by calculation from these basic measurements. Properties that can be aspects of luminaire testing include:

Photometric properties:

- Total flux
- Bare lamp output (relative photometry only; see **Section 14.3**)
- Intensity distribution
- Luminances

Derived photometric properties:

- Zonal lumens
- Luminous efficiency
- Coefficients of utilization
- Spacing criteria
- Beam and field flux
- Beam and field angles
- Various luminaire classifications

Other measured or tested properties:

- Thermal performance
- Construction
- Water and vapor sealing
- Air-handling performance

14.5 Luminaire Photometric Testing

Luminaire photometric testing is usually intended to describe the photometric (or radiometric) performance characteristics of a typical individual luminaire from a relatively large population of commercially produced luminaires of a single type. Though it is ideal to photometrically test a sample of luminaires of a given type, this is rarely done. Luminaires submitted for photometric testing should be representative, but as a practical matter there is no way to be certain of this, other than extensive testing. Nevertheless, commercial luminaire photometry usually provides a useful representation of how a large number of the same type of luminaire will perform. Most luminaire types used in architectural lighting have an appropriate standard governing the equipment, procedures, test conditions, instrumentation, and test report contents to be used or provided in luminaire photometric testing. Standards available include guidance for testing these types of luminaires:

- Indoor fluorescent⁵⁷
- Outdoor fluorescent⁵⁸
- Indoor high intensity discharge (HID)⁵⁹
- Indoor incandescent⁵⁶
- Roadway⁶⁰
- Floodlights⁶¹
- Searchlights⁶²
- Light emitting diode (LED)¹⁷

14.6 Derived Photometric Characteristics

Luminaire photometric reports usually contain performance data that are not measured directly, but calculated by standard procedures and listed as part of the photometric report. Several of those derived properties (see list in **Section 14.4**) are described here.

14.6.1 Zonal Lumens. Zonal lumens describe the flux distribution of the luminaire using solid angle elements sized and shaped to the needs of the typical application for the luminaire and the coordinate system used for distribution photometry. Standards governing luminaire photometry define these sizes and shapes. Generally, the lumens in a zone, Φ_{zone} , are calculated from:

$$\Phi_{\text{zone}} = \int_{\Omega_{\text{zone}}} I(d\omega) d\omega, \quad (14-3)$$

where:

Ω_{zone} = solid angle of the zone, in steradians

$d\Omega$ = differential element of solid angle in the zone

$I(d\Omega)$ = luminous intensity in the direction of $d\Omega$, in candelas

Generally, the lumens, Φ_{zone} , are usually approximated by:

$$\Phi_{\text{zone}} = \bar{I} \Omega_{\text{zone}}, \quad (14-4)$$

where:

Ω_{zone} = solid angle of the zone, in steradians

$\bar{I}$ = appropriate average luminous intensity over the zone

14.6.2 Luminaire Efficiency and Photometric Efficiency

Luminaire efficiency is usually defined as the ratio of lumens emitted by the luminaire to the lumens emitted by the lamps. If relative photometry is used, then the lumens emitted by the lamps are, by definition, rated lumens. In some cases of relative photometry, lamp output is assumed to be 1,000 lm.

When relative photometric testing procedures are followed to determine rated lumens with the lamps outside the luminaire, the bulb wall temperature may be significantly below rated bulb wall temperature. Consequently, during this determination the lumen output may be lower than rated lumens and lower than when the lamps are operating in the luminaire. This can produce a ratio of luminaire lumens to procedure-determined rated lamp lumens greater than 1.0.⁶³ Thus, it is sometimes necessary to determine and use a luminaire thermal factor, and the total photometric efficiency of the luminaire is the luminaire efficiency times the luminaire thermal factor.

14.6.3 Average Luminance. Depending on the form of the luminaire being tested, determining the apparent area to be used in the calculation of average luminaire luminance may require special attention to calculate and may not be simply a planar luminous aperture under foreshortening.⁶⁴

14.6.4 Beam Type and Characterization. Based on the intensity distribution and the subsequent determination of zonal lumens, the test luminaire can be assigned a classification for outdoor applications.⁶⁵

14.6.5 Coefficients of Utilization.

Indoor:

Coefficients of utilization for indoor luminaires are performance indicators used to predict average illuminances produced by luminaires typically arranged in uniform arrays. (Refer to *ANSI/IES LS-6-20, Lighting Science: Calculation of Light and Its Effects* for details concerning the determination of these coefficients.⁶⁶)

Roadway:

Coefficients of utilization are calculated for roadway luminaires for use in determining the average illuminance on a roadway. The calculations that use these values usually assume a uniform pole spacing.⁶⁷

15.0 Field Measurements

Evaluating a lighting installation in the field usually involves illuminance and luminance measurements at the site of the installation. The purpose of such measurements can be to:

- Validate design calculations
- Isolate problems in, and apparent differences between, expected and observed illuminance
- Complete a post-occupancy evaluation
- Assess an existing installation in advance of upgrading or retrofitting lighting equipment
- Provide a benchmark for renovation or expansion
- Determine compliance with specifications or codes
- Reveal the need for maintenance, modification, or replacement

The purpose usually determines the type and extent of the measurements.

These evaluations usually take the form of a survey with illuminance (and, less frequently, luminance) being measured at enough carefully chosen positions in an installation to reliably determine averages and assess typical maximum and minimum values. Less frequently, measurements are required at essentially a single location. Good practice requires recording a complete, detailed description of the surveyed area and all factors that might affect results, such as interior surface reflectances, presence of daylight from sky and/or sun, ambient temperature, spot temperature of lamp and/or luminaire, stabilization time, presence of objects in the space, light source type and age, voltage, and instruments used in the survey.

Precautions and preparations common to most measurement surveys include:

- Using illuminance meters with adequate cosine and spectral correction
- Conducting measurements at a temperature that is above 15 °C (60 °F) and below 50 °C (120 °F)
- Avoidance of casting shadows on the detector
- Standing far enough away from the detector, especially when wearing light-colored clothes, to

prevent light from the source from being reflected onto it

- In the case of a fluorescent or HID system, ensuring that the system is lighted for at least 1 hour before making measurements
- Ensuring that light sources are seasoned before the survey is made: at least 100 hours of operation for gaseous sources (fluorescent, HID, LPS), and 20 hours or less for common sizes of filament lamps and LED light sources

15.1 Interior Measurements

Measurements in interior spaces are usually made for the purpose of evaluating an existing condition or to verify the performance of a new lighting installation. Average illuminance over large open areas is determined from measurements made at selected points.

Illuminance at a point or a small neighborhood of points is usually measured at specific task areas. Luminance measurements in interiors, though less common, are made to verify design ratios or investigate conditions that may lead to glare or veiling reflections.

15.1.1 Average Illuminance.

Determination of Average Illuminance on a Horizontal Plane:

The measuring instrument should be positioned so that when readings are taken, the surface of the detector is in a horizontal plane and 760 mm (about 30 in.) above the floor, or at whatever height is of interest. The area should be divided into approximately equal-sized squares, taking a reading in the center of each square and calculating the arithmetic mean. A measurement grid of 0.6 m (2 ft) is suitable for many spaces. Using a relatively dense rectangular grid of measurement locations is usually necessary in spaces that are obstructed, lack orthogonal geometry, or have highly uneven illumination. For spaces with unusual room cavity ratios or highly uneven illumination, as in corridors under emergency lighting conditions, a denser grid of measurement points may be necessary.

For more-uniform and symmetric rooms and luminaire positions, a uniform survey method for measuring and reporting the necessary data for interior applications

as been developed.⁶⁸ The method has been found generally reliable to within an accuracy of 10 percent. It has the advantage of using a weighted average of measurements made at select locations to minimize the number of measurements required. The room types suited for this survey method are shown in **Figure 15-1**.

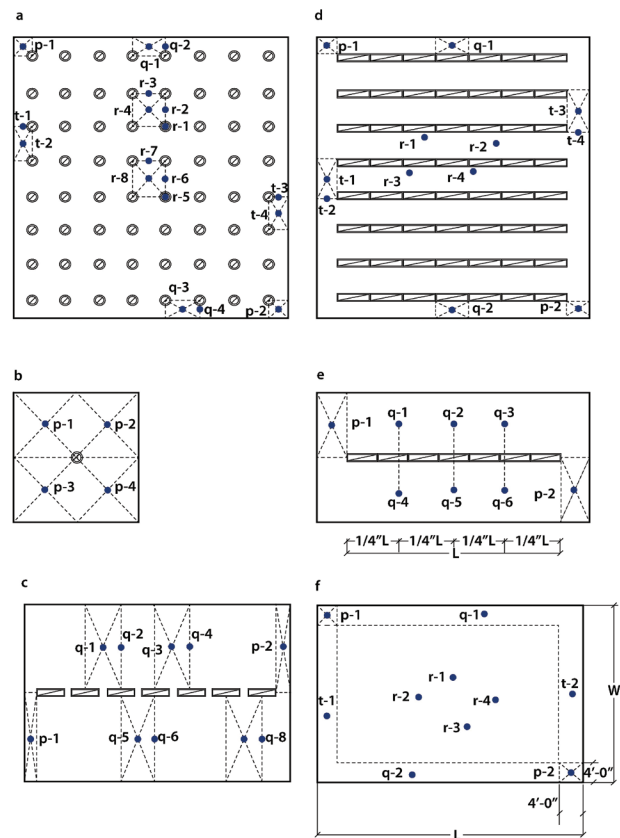


Figure 15-1. IES survey measurement locations for various rooms: a) regular area with symmetrically located luminaires; b) regular area with symmetrically located single luminaire; c) regular area with single row of continuous luminaires; d) regular area with two or more continuous rows of luminaires; e) regular area with single row of continuous luminaires; f) regular area with uniform indirect lighting. (© Illuminating Engineering Society)

Regular Area with Symmetrically Spaced Luminaires in Two or More Rows:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1a**) can be determined from:

$$\bar{E} = \frac{R(N-1)(M-1) + Q(N-1) + T(M-1) + P}{NM}, \quad (15-1)$$

where:

N = Number of luminaires per row

M = Number of rows

R = Average of measurements at stations r-1, r-2, r-3, and r-4 for a typical inner bay; and at stations r-5, r-6, r-7, and r-8 for a typical centrally located bay

Q = Average of measurement at stations q-1, q-2, q-3, and q-4 in two typical half-bays on each side of the room

T = Average of measurements at stations t-1, t-2, t-3, and t-4 in two typical half-bays at each end of the room

P = Average of measurements at stations p-1 and p-2 in two typical corner quarter-bays

Regular Area with Symmetrically Located Single Luminaire:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1b**) can be determined from:

$$\bar{E} = P, \quad (15-2)$$

where:

P = Average of measurements at stations p-1, p-2, p-3, and p-4 in all four quarter-bays

Regular Area with Single Row of Individual Luminaires:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1c**) can be determined from:

$$\bar{E} = \frac{Q(N-1) + P}{N}, \quad (15-3)$$

where:

N = Number of luminaires

Q = Average of measurements at stations q-1 through q-8 in four typical half-bays, located two on each side of the area

P = Average of measurements at stations p-1 and p-2 for two typical corner quarter-bays

Regular Area with Two or More Continuous Rows of Luminaires:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1d**) can be determined from:

$$\bar{E} = \frac{R(N-1)(M-1) + QN + T(M-1) + P}{M(N-1)}, \quad (15-4)$$

where:

N = Number of luminaires per row

M = number of rows

R = Average of measurements at stations r-1 through r-4, located near the center of the area

Q = Average of measurements at stations q-1 and q-2, located at the middle of each side of the room, and midway between the outside row of luminaires and the wall

T = Average of measurements at stations t-1 through t-4 at each end of the room

P = Average of measurements at stations p-1 and p-2 in two typical corners

Regular Area with Single Row of Continuous Luminaires:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1e**) can be determined from:

$$\bar{E} = \frac{Q(N-1) + P}{N}, \quad (15-5)$$

where:

N = Number of luminaires

Q = Average of measurements at stations q-1 through q-6

P = Average of measurements at stations p-1 and p-2 in typical corners

Regular Area with Uniform Indirect Lighting:

The average illuminance, $\bar{E}$, in such a space (see **Figure 15-1f**) can be determined from:

$$\bar{E} = \frac{R(L-8)(W-8) + 8Q(L-8) + 8T(W-8) + 64P}{WL}, \quad (15-6)$$

where:

W = Number of luminaires per row

L = Number of rows

R = Average of measurements at stations r-1 through r-4, located at random in the central portion of the area

Q = Average of measurements at stations q-1 and q-2, located 0.6 m (2 ft) from the long walls and at random lengthwise in the room

T = Average of measurements at stations t-1 and t-2, located 0.6 m (2 ft) from the short walls and at random crosswise in the room

P = Average of measurements at stations p-1 and p-2, located at diagonally opposite corners and 0.6 m (2 ft) from each wall

15.1.2 Illuminance at a Point. With task, general, and supplementary lighting in use, the illuminance at the point of work should be measured with the worker in his or her normal working position. It is important to note that this will generally not correspond to an illuminance prediction at that point, since body shadow is rarely taken into account in illuminance calculations. The measuring instrument should be located so that when readings are taken, the surface of the light-sensitive cell is in the plane of the visual task or of that portion of the visual task on which the critical visual processing is required—horizontal, vertical, or inclined.

15.1.3 Luminance. Luminance surveys should be made under actual working conditions with measurements at specified work point locations and with the combinations of daylight and electric lighting facilities available. Consideration should be given to sun position and weather conditions, both of which can have a marked effect on the luminance distribution. All lighting in the area—task, general, and supplementary—should be in normal use. Work areas used only in the daytime should be surveyed in the daytime; work areas used both day and night should be surveyed under both conditions, as the luminance distribution and the possibilities of comfort and discomfort can differ markedly between them. Nighttime surveys should be made at night or with shades drawn. Daytime surveys should be made with shades adjusted to positions actually set by the occupants. Measurement locations are usually those defined by luminance ratios or limits that have been specified. (Refer to the appropriate IES Recommended Practice [RP] documents for recommended luminance ratios and limits.)

15.2 Outdoor Measurements

For an accurate evaluation of roadway and many floodlight installations, illuminance measurements need to be made with particular care, especially regarding the level of the illuminance meter or its alignment with the intended illuminance measurement plane. Typical preparation for an illuminance survey consists of these steps:

- Inspect and record the condition of the luminaires (e.g., globes, reflectors, refractors, lamp positioning).

- In the case of roadway lighting, make sure the luminaires are level and their vertical and lateral placement is as designed.
- Unless the purpose of the test is to check depreciation or actual in-service performance, all units should be cleaned. In cases where light sources are replaceable (e.g., luminaires with socketed lamps), new lamps should be installed. In cases, where the light sources are integral to the luminaires (e.g., integral-LED luminaires) the sources are generally intended for the life of the luminaire and should be inspected for unusually dim or non-operable LEDs and LED chains but not otherwise replaced; such action should be recorded as part of the measurement records.
- Any new light sources should be seasoned properly according to the seasoning practice for the light source technology. While inoperative light sources may be readily noticed in some roadway installations (e.g., single-lamp HID luminaires), they can easily be overlooked in large floodlighting systems and other multi-source luminaires and lighting system installations. If these inoperative light sources are not replaced for the field survey, it is important that proper consideration be given when evaluating the results.
- Measure and record the mounting heights of the luminaires.
- Measure and record the locations of the poles, the number of luminaires per pole, the wattage of the luminaires or light sources, and other pertinent data. Check these data against the recommended layout; a small change in the location or adjustment of the luminaires can make a considerable difference in illuminance.
- Determine and record the burning hours of the installed light sources.
- Consider the impact of stray light on the measurements.
 - The survey should be made only when the atmosphere is reasonably clear.
 - Extraneous light produced by a store, a service station, or other lights in the vicinity requires careful attention in outdoor lighting tests.

- Luminaire voltage should be measured because it can affect illuminance with some light source and electrical gear technologies. At night, during the hours when the luminaires are normally used, record the voltage at the supply voltage input point for the luminaires (e.g., a supply-connection terminal block internal to the luminaire).
 - The voltage at the main switch can be used instead, provided allowance is made for the voltage drop to the individual luminaires.
 - If discharge lamps are used, record the input voltage to the ballast at the ballast terminals. Discharge lamps should be operated at least 60 minutes to reach normal operating conditions before measurements are made.

Measurements should be made with a recently calibrated illuminance meter with sufficiently accurate spectral and spatial response, and which is capable of being leveled for horizontal measurements or positioned accurately for other measurement planes as required. The photometer should be selected for its portability, repeatability, and measurement range.

- For roadway lighting systems, at least one traffic lane will need to be closed for substantial periods. Because of this difficulty and the expense of making field measurements of pavement luminance, it is common to use a computerized design procedure using point calculations of horizontal illuminance level at each of the pavement luminance measurement points recommended. As a check on the computer calculations, it is necessary only to measure the illuminance at a reduced number of points.⁶⁷
- For roadway signs, the minimum and maximum illuminance levels are determined by scanning the sign face. Additional illuminance measurements are taken at specific locations according to the sign size. Luminance measurements are also made for both externally and internally illuminated signs.⁶⁹
- For sports installations, the sports area (or the portion of the area under immediate consideration) should be divided into test areas of approximately 5 percent of the total area, and readings should be taken at the center of each area.⁷⁰ Some specific measurement grids have been developed

for particular types of playing fields and are recommended for field measurements in these applications.

- So-called “TV-illuminance” requires the illuminance meter to be oriented so that its measurement normal points to the camera position(s).
- Readings should be made at each test station, with repeat measurements at the first station frequently enough to ensure stability of the system and repeatability of results. Readings should be reproducible within 5 percent. Enough readings should be taken such that additional readings in similar locations will not change the average results significantly.

Many outdoor lighting applications have an appropriate standard governing the equipment, procedures, test conditions, instrumentation, and test report contents to be used for field tests. These include:

Outdoor HID⁷¹

Parking areas⁶⁹

Roadway⁶⁹

Roadway signs⁷⁰

Tunnel⁶⁹

Area and sports⁷¹

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